A DISTINGUISHED TRIANGLE ON AN OPEN LIFTS TO A COMPATIBLE SYSTEM OF DISTINGUISHED TRIANGLES ON THE AMBIENT SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G4M (sources/stacks-duality/pinned-original-source.tex line 8145).

Allowed standard prerequisites (editorial).

- Existence of Deligne systems and of lifts of bounded coherent objects from an open. Same chapter, Lemma labelled lemma-extension-by-zero part (2), cited at proof lines 8161-8163. Provides (K_n) and L'.
- Surjectivity of Hom onto the open for a Deligne system. Same chapter, Lemma labelled lemma-lift-map, cited at proof lines 8163-8166. Lifts the arrow K -> L to a morphism over X.
- The bounded coherent derived category as the derived category of the abelian category of coherent modules. Derived Categories of Schemes, Proposition labelled perfect-proposition-DCoh, cited at proof lines 8176-8178. Represents the lifted morphism by a morphism of complexes.
- Cones of complexes and axiom TR3. Derived Categories, Section labelled derived-section-cones and Lemma labelled derived-lemma-third-isomorphism-triangle, cited at proof lines 8188-8200. Show that the cones form the required compatible triangles.

Source reference. Original lemma 0G4L: sources/stacks-duality/pinned-original-source.tex, source lines 8103–8103. Same-chapter pointer into the author environment lemma at source line 8103; invoked at proof line 8162 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G2H: sources/stacks-duality/pinned-original-source.tex, source lines 7920–7920. Same-chapter pointer into the author environment lemma at source line 7920; invoked at proof line 8165 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

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Source reference. Original proposition 0FDB: sources/stacks-chow/tags.txt, source lines 19211–19211. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8172 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 014D: sources/stacks-chow/tags.txt, source lines 1468–1468. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8185 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 014Z: sources/stacks-chow/tags.txt, source lines 1490–1490. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8193 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 014A: sources/stacks-chow/tags.txt, source lines 1465–1465. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8199 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $j : U \to X$ be an open immersion of Noetherian schemes. Let*

$$K \to L \to M \to K[1]$$

be a distinguished triangle of $D^b_{Coh}(\mathcal{O}_U)$. Then there exists an inverse system of distinguished triangles

$$K_n \to L_n \to M_n \to K_n[1]$$

in $D^b_{Coh}(\mathcal{O}_X)$ such that (K_n), (L_n), (M_n) are Deligne systems and such that the restriction of these distinguished triangles to U is isomorphic to the distinguished triangle we started out with.

Proof. Let (K_n) be as in Lemma 0G4L part (2). Choose an object L' of $D^b_{Coh}(\mathcal{O}_X)$ whose restriction to U is L (we can do this as the lemma shows). By Lemma 0G2H we can find an n and a morphism $K_n \to L'$ on X whose restriction to U is the given arrow $K \to L$. We conclude there is a morphism $K' \to L'$ of $D^b_{Coh}(\mathcal{O}_X)$ whose restriction to U is the given arrow $K \to L$.

By Derived Categories of Schemes, Proposition 0FDB we can find a morphism $\alpha^\bullet : \mathcal{F}^\bullet \to \mathcal{G}^\bullet$ of bounded complexes of coherent $\mathcal{O}_X$-modules representing $K' \to L'$. Choose a quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$ whose vanishing locus is $X \setminus U$. Then we let $K_n = \mathcal{I}^n\mathcal{F}^\bullet$ and $L_n = \mathcal{I}^n\mathcal{G}^\bullet$. Observe that $\alpha^\bullet$ induces a morphism of complexes $\alpha^\bullet_n : \mathcal{I}^n\mathcal{F}^\bullet \to \mathcal{I}^n\mathcal{G}^\bullet$. From the construction of cones in Derived Categories, Section 014D it is clear that

$$C(\alpha_n)^\bullet = \mathcal{I}^n C(\alpha^\bullet)$$

and hence we can set $M_n = C(\alpha_n)^\bullet$. Namely, we have a compatible system of distinguished triangles (see discussion in Derived Categories, Section 014Z)

$$K_n \to L_n \to M_n \to K_n[1]$$

whose restriction to U is isomorphic to the distinguished triangle we started out with by axiom TR3 and Derived Categories, Lemma 014A. □